

A Weak- ℓ_1 Solution of the B_1^n Linear Part 1' Step

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Status

This packet gives a full solution candidate, likely valid pending expert review, for the linear Part 1' separation-dimension-reduction challenge in Mendelson–Milman–Paouris [1] in the case $K = B_1^n$. The target body after dimension reduction is a fixed multiple of the weak- ℓ_1 star body. The result upgrades the earlier ordinary- ℓ_1 Cauchy packet: the ordinary- ℓ_1 body incurred an $O(k \log k)$ separation loss, while the weak- ℓ_1 body gives a universal constant loss.

The theorem below solves the linear Part 1' step, not the entire generalized dual Sudakov program. To run the full program with this target family one would still need the corresponding Part 2 weak Sudakov estimate for weak- ℓ_1 star bodies.

Source Challenge

The source paper is Shahar Mendelson, Emanuel Milman, and Grigoris Paouris, *Generalized Dual Sudakov Minoration via Dimension Reduction - A Program*, arXiv:1610.09287. The paper formulates a linear version of Part 1: given e^k many K -separated points and a measure μ with $\mu(K) \geq 1/e$, find a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $m \leq Ak$, and a massive star body $L \subset \mathbb{R}^m$ such that the images remain $(1/R)L$ -separated. In the conclusion, the authors identify Part 1 as the main remaining challenge and explicitly say that it is significant even for $K = B_1^n$.

μ is assumed 1-pure (e.g. super-Gaussian, sub-Gaussian or unconditional), then by Proposition 8.5, so will be all of its marginals $\nu \in \mathcal{M}_m$, for which we have a Weak Sudakov Minoration result by Theorem 9.1. In particular, up to the Slicing Problem, Part 2 is completely established.

Consequently, it is clear that the main remaining challenge in completing The Program lies in establishing Part 1 of The Program. This is a significant challenge even

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for some specific convex bodies K besides ellipsoids, such as for $K = B_1^n$. To carry out this Separation Dimension-Reduction step, it seems that we would need to employ other measures on the Grassmannian $G_{n,m}$ besides the uniform Haar measure, upon which most of the (Euclidean) Asymptotic Geometric Analysis theory is built. In our opinion, this is a fascinating challenge, which we plan to explore in a future work.

Definitions

For $y \in \mathbb{R}^m$, define the weak- ℓ_1 quasi-norm

$$\|y\|_{1,\infty} = \sup_{s>0} s \#\{j : |y_j| > s\}.$$

Equivalently, if $y_1^* \geq \dots \geq y_m^*$ is the non-increasing rearrangement of $(|y_j|)$, then

$$\|y\|_{1,\infty} = \max_{1 \leq r \leq m} r y_r^*.$$

Let

$$W_m = \{y \in \mathbb{R}^m : \|y\|_{1,\infty} \leq 1\}.$$

This is an origin-symmetric compact star body with non-empty interior.

We use the source paper's convention that K -separated points are points whose translates by K are mutually disjoint. For $K = B_1^n$, this implies $\|x_i - x_j\|_1 \geq 2$ up to the immaterial boundary convention; the proof below only uses the weaker lower bound $\|x_i - x_j\|_1 \geq 1$.

Proof Intuition

The Cauchy law is exactly 1-stable: a row of a Cauchy matrix applied to x has scale $\|x\|_1$. In the earlier ordinary- ℓ_1 attempt, the vector of m Cauchy variables had typical ℓ_1 -size $m \log m$, which produced the unwanted $O(k \log k)$ loss.

The right scale for Cauchy tails is weak ℓ_1 . If $G = (G_1, \dots, G_m)$ has independent standard Cauchy coordinates, then $\|G\|_{1,\infty} = O(m)$ with fixed probability. After scaling the random map by $1/m$, the image of every point of B_1^n falls in a fixed multiple of W_m with fixed probability, yielding massiveness by averaging. At the same time, the event $\|G\|_{1,\infty} \leq \eta m$ has exponentially small probability for small η , because it would force far fewer than the typical number of coordinates to exceed 1. This exponential small-ball bound is strong enough to union-bound over e^k point pairs in dimension $m \simeq k$.

The Result

Theorem 1 (Weak- ℓ_1 Part 1' for B_1^n). *There are universal constants $A_0, A_1, R_0, q_0, K_0 > 0$ with $q_0 \geq 1$ such that the following holds. Let $k \geq K_0$, let $M \leq e^k$, and let $x_1, \dots, x_M \in \mathbb{R}^n$ satisfy*

$$\|x_i - x_j\|_1 \geq 1 \quad (i \neq j).$$

Let μ be a Borel probability measure on $\mathbb{R}^n$ with $\mu(B_1^n) \geq 1/e$. Then there are $m \leq A_1 k$, a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and

$$L = A_0 W_m$$

such that

$$Tx_i - Tx_j \notin \frac{1}{R_0} L \quad (i \neq j)$$

and

$$T_*\mu(L) \geq e^{-q_0}.$$

In particular, this verifies the linear Part 1' separation-dimension-reduction step for $K = B_1^n$, with universal constants, using the reduced target family $L = A_0 W_m$.

Lemma 2 (Weak- ℓ_1 estimates for Cauchy vectors). *Let $G_1, \dots, G_m$ be independent standard Cauchy random variables and set $G = (G_1, \dots, G_m)$. There are universal constants $\eta, A, c > 0$ such that, for every $m \geq 1$,*

$$\mathbb{P}\{\|G\|_{1,\infty} \leq \eta m\} \leq e^{-cm}$$

and

$$\mathbb{P}\{\|G\|_{1,\infty} \leq Am\} \geq \frac{3}{4}.$$

Proof. For the lower-tail estimate, use $\mathbb{P}\{|G_1| > 1\} = 1/2$. If $\|G\|_{1,\infty} \leq \eta m$, then

$$\#\{j : |G_j| > 1\} \leq \eta m.$$

Taking $\eta > 0$ sufficiently small, the Chernoff bound for a binomial $\text{Bin}(m, 1/2)$ variable gives

$$\mathbb{P}\{\|G\|_{1,\infty} \leq \eta m\} \leq e^{-cm}.$$

For the upper-tail estimate, write

$$N(u) = \#\{j : |G_j| > u\}.$$

There is a universal C_0 such that $\mathbb{P}\{|G_1| > u\} \leq C_0/u$ for all $u > 0$. If $\|G\|_{1,\infty} > Am$, then for some $u > A$ one has $uN(u) > Am$. Choose $r \geq 0$ with $2^r A < u \leq 2^{r+1} A$. Then

$$N(2^r A) > \frac{m}{2^{r+1}}.$$

For those r with $m/2^{r+1} \geq 1$, the binomial tail bound gives

$$\mathbb{P}\left\{N(2^r A) > \frac{m}{2^{r+1}}\right\} \leq \left(\frac{2eC_0}{A}\right)^{m/2^{r+1}},$$

provided A is larger than a fixed multiple of C_0 . Summing over these dyadic r 's yields at most

$$\sum_{s=1,2,4,\dots} \left(\frac{2eC_0}{A}\right)^s.$$

The remaining dyadic levels have $2^r A \gtrsim Am$, and their total contribution is bounded by

$$\mathbb{P}\{\max_j |G_j| \gtrsim Am\} \leq \frac{C_1}{A}.$$

Choosing A sufficiently large makes the total probability at most $1/4$. □

Proof of Theorem 1. Let $m = \lceil C_2 k \rceil$, where C_2 will be chosen as a sufficiently large universal constant. Let $(g_{jr})_{1 \leq j \leq m, 1 \leq r \leq n}$ be independent standard Cauchy variables and define

$$(Tx)_j = \frac{1}{m} \sum_{r=1}^n g_{jr} x_r, \quad j = 1, \dots, m.$$

The standard 1-stability identity for Cauchy variables gives, for each fixed $x \in \mathbb{R}^n$,

$$Tx \stackrel{d}{=} \frac{\|x\|_1}{m} (G_1, \dots, G_m),$$

with independent standard Cauchy G_j 's.

Let $h = x_i - x_j$. Since $\|h\|_1 \geq 1$, Lemma 2 implies

$$\mathbb{P}\{\|Th\|_{1,\infty} \leq \eta\} \leq \mathbb{P}\{\|G\|_{1,\infty} \leq \eta m\} \leq e^{-cm}.$$

Therefore

$$\mathbb{P}\{\exists i < j : \|T(x_i - x_j)\|_{1,\infty} \leq \eta\} \leq M^2 e^{-cm} \leq \exp(2k - cm).$$

Choose C_2 so large that this probability is at most e^{-3} .

Now set $L = AW_m$, where A is the constant in Lemma 2. For every $x \in B_1^n$, the same stability identity and the upper-tail estimate give

$$\mathbb{P}\{Tx \in L\} \geq \frac{3}{4}.$$

Consequently,

$$\mathbb{E}[T_*\mu(L)] = \int \mathbb{P}\{Tx \in L\} d\mu(x) \geq \frac{3}{4} \mu(B_1^n) \geq \frac{3}{4e}.$$

Since $0 \leq T_*\mu(L) \leq 1$, it follows that

$$\mathbb{P}\left\{T_*\mu(L) \geq \frac{3}{8e}\right\} \geq \frac{3}{8e}.$$

After increasing C_2 if needed, the separation failure probability is smaller than $3/(8e)$. Hence there exists a realization of T for which both events hold:

$$\|T(x_i - x_j)\|_{1,\infty} > \eta \quad (i \neq j), \quad T_*\mu(AW_m) \geq \frac{3}{8e}.$$

Put $A_0 = A$, $R_0 = A/\eta$, and

$$q_0 = \log\left(\frac{8e}{3}\right).$$

Then $T_*\mu(L) \geq e^{-q_0}$, and

$$\eta W_m = \frac{\eta}{A} L = \frac{1}{R_0} L,$$

so the images are $(1/R_0)L$ -separated. Finally, for $K_0 \geq 2q_0$ we have $q_0 \leq k/2$, exactly as required in Part 1'. $\square$

Verification and Novelty Check

The proof uses only the elementary 1-stability of Cauchy variables, Chernoff bounds for binomial variables, and a dyadic weak- ℓ_1 tail estimate. No computational verification is required.

Local run indexes were searched for arXiv:1610.09287, “Cauchy”, “weak ℓ_1 ”, “weak- ℓ_1 ”, “Part 1”, and “one-sided Johnson–Lindenstrauss”. They contained the earlier ordinary ℓ_1 partial packet but no weak- ℓ_1 full Part 1’ packet. External web searches on July 3, 2026 for combinations of “Cauchy weak ℓ_1 dimension reduction”, “one-sided Johnson–Lindenstrauss Cauchy weak ℓ_1 ”, and “B1 weak ℓ_1 Sudakov dimension reduction” did not surface a direct match. The arXiv abstract for [1] records that the paper establishes the separation step for ellipsoids and, with logarithmic losses, for cubes; it does not state the weak- ℓ_1 B_1^n result proved here.

Limitations and Human Review Recommendation

This packet should be reviewed as a full solution of the linear Part 1' step for $K = B_1^n$. It should not be cited as a full proof of the generalized dual Sudakov conjecture, because Part 2 has not been proved for the weak- ℓ_1 target family AW_m .

The most important checks are:

- the dyadic proof that $\|G\|_{1,\infty} = O(m)$ with constant probability;
- the lower-tail estimate $\mathbb{P}\{\|G\|_{1,\infty} \leq \eta m\} \leq e^{-cm}$;
- the translation from $\|T(x_i - x_j)\|_{1,\infty} > \eta$ to $(1/R_0)L$ -separation.

References

- [1] S. Mendelson, E. Milman, and G. Paouris, *Generalized Dual Sudakov Minoration via Dimension Reduction - A Program*, arXiv:1610.09287, 2016; revised 2018.